

PROJECT REPORT

UniSwap – Campus Marketplace Web Application

Submitted by

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Certificate

(To be signed by your Guide, HOD, and Principal as per your college format.)

Acknowledgement

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Abstract

UniSwap is a web-based campus marketplace developed to simplify buying and selling among students and staff within a college campus. Existing public marketplaces often lack

campus-specific trust, making transactions less convenient. UniSwap addresses this challenge by providing a secure platform where verified users can list products, browse available items, communicate through real-time chat, and manage transactions efficiently.

The application is built using **React.js**, **Vite**, **HTML5**, **CSS3**, and **JavaScript** for the frontend, while **Supabase (PostgreSQL)** is used as the backend database with **Supabase Realtime** enabling instant messaging. The platform supports authentication, product management, dashboards, notifications, wishlist functionality, and an admin approval workflow. The modular architecture ensures scalability and maintainability, making UniSwap suitable for future enhancements such as AI-powered recommendations, online payments, and mobile application support.

Keywords: Campus Marketplace, React.js, Supabase, Realtime Chat, E-Commerce, Web Application.

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Chapter 1 – Introduction

1.1 Introduction

With the increasing adoption of online marketplaces, students and staff require a dedicated platform to exchange books, electronics, hostel essentials, furniture, and other items within the campus community. Existing platforms are designed for the general public and often involve issues such as long-distance travel, unreliable sellers, and lack of institutional trust.

UniSwap is designed specifically for educational institutions, allowing verified campus members to buy and sell products in a secure environment. By integrating product listings, real-time communication, and user dashboards into a single platform, UniSwap enhances convenience while promoting sustainable reuse of resources within the campus.

Chapter 2 – Problem Statement

Students frequently face challenges in buying and selling used items within the campus. Public marketplaces lack verification mechanisms for campus users, resulting in trust issues and communication delays. There is no centralized platform that enables students and staff to exchange products efficiently within the institution.

Chapter 3 – Objectives

Primary Objective

Develop a secure web application that enables students and staff to buy and sell products within the campus.

Secondary Objectives

- Provide secure authentication.
 - Allow users to list and manage products.
 - Support real-time buyer–seller communication.
 - Enable product search and category filtering.
 - Provide wishlist and notification features.
 - Support product approval before publishing.
-

Chapter 4 – Existing System

Current alternatives include:

- OLX
- Facebook Marketplace
- WhatsApp Groups

Limitations

- No campus verification.
 - Trust and safety concerns.
 - No centralized campus marketplace.
 - Difficult product discovery.
 - Limited organization of listings.
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Chapter 5 – Proposed System

UniSwap provides a dedicated campus marketplace with:

- Secure user authentication.
 - Product listing and management.
 - Search and filtering.
 - Real-time chat.
 - Notifications.
 - Wishlist.
 - Admin approval workflow.
 - Responsive and user-friendly interface.
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Chapter 6 – System Requirements

Software Requirements

- React.js
- Vite
- HTML5
- CSS3
- JavaScript
- Supabase (PostgreSQL)
- Git & GitHub
- Visual Studio Code

Hardware Requirements

Minimum

- Intel Core i3
- 8 GB RAM
- 10 GB free storage
- Internet connection

Recommended

- Intel Core i5 or above
 - 16 GB RAM
 - SSD
 - Stable internet connection
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Chapter 7 – Feasibility Study

Technical Feasibility

The selected technologies are open-source, widely supported, and suitable for developing a scalable web application.

Economic Feasibility

The project uses free development tools and Supabase's free tier, minimizing development costs.

Operational Feasibility

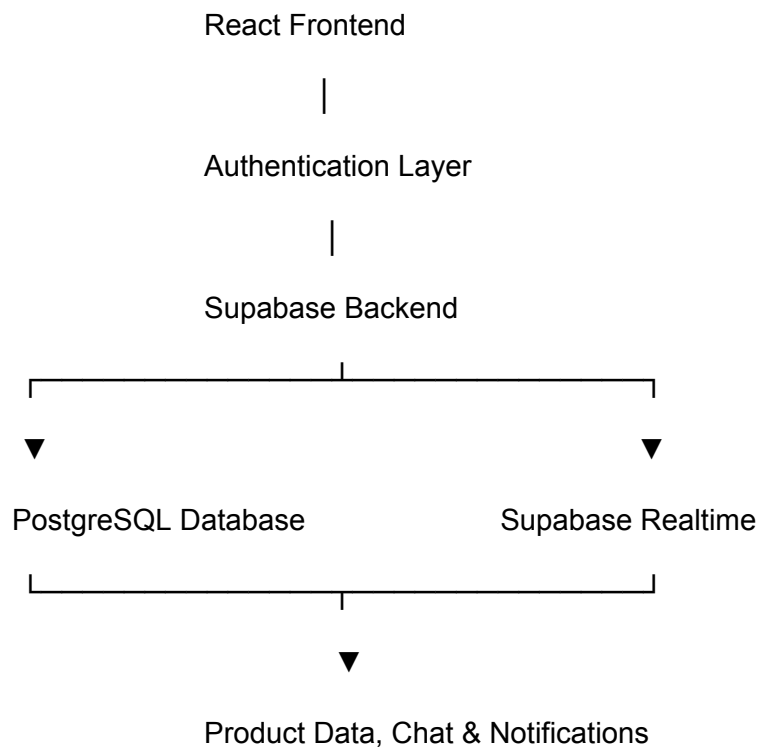
The application is easy to use and can be adopted by students and staff with minimal training.

Chapter 8 – System Architecture

Users

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Chapter 9 – Module Description

Authentication Module

- Login
- Create Account
- Forgot Password
- Session Management

Home Module

- Product feed
- Categories
- Search
- Filters

Seller Module

- Add product
- Upload images
- Optional video
- Edit listing

- Delete listing
- Approval status

Buyer Module

- Browse products
- View product details
- Wishlist
- Chat with seller
- Make offer

Chat Module

- One-to-one messaging
- Realtime updates
- Typing indicator
- Notifications
- Conversation history

Dashboard Module

- Profile management
 - My listings
 - Wishlist
 - Notifications
 - Statistics
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Chapter 10 – Database Design

Main Tables

- Users
- Products
- Product Images
- Product Videos
- Chat Threads
- Chat Messages
- Notifications
- Wishlist
- Product Approval

Explain the primary keys, foreign keys, and relationships. Include your ER diagram here.

Chapter 11 – Technology Stack

Layer	Technology
Frontend	React.js
Build Tool	Vite
Styling	HTML5, CSS3
Programming	JavaScript (ES6+)
Database	PostgreSQL (Supabase)
Backend Services	Supabase
Realtime	Supabase Realtime
Version Control	Git, GitHub

Chapter 12 – Implementation

Describe the implementation of each module:

- Authentication flow.
- Product listing process.

- Product approval workflow.
- Dashboard features.
- Chat implementation using Supabase Realtime.
- Notification system.
- Search and filtering.
- Responsive UI.

Include screenshots of:

- Login page
 - Registration page
 - Home page
 - Product details
 - Sell product page
 - Dashboard
 - Chat interface
 - Notifications
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Chapter 13 – Testing

Types of Testing

- Unit Testing
- Integration Testing
- Functional Testing
- User Interface Testing

Sample Test Cases

Test Case	Expected Result	Status
User Login	Login successful	Pass
Create Account	User registered	Pass
Add Product	Product saved	Pass

Search	Correct results	Pass
Product		

Chat Message	Delivered in real time	Pass
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Wishlist	Product added	Pass
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Chapter 14 – Results

The application successfully provides a complete campus marketplace where students and staff can register, list products, search for items, communicate through real-time chat, and manage transactions. The responsive interface and modular architecture ensure smooth operation across devices, while the admin approval workflow enhances trust and security within the marketplace.

Chapter 15 – Future Enhancements

- AI-based product recommendations.
 - AI chatbot for user assistance.
 - Product rating and review system.
 - Online payment gateway integration.
 - QR-code verification for transactions.
 - Email and push notifications.
 - Mobile application (Android/iOS).
 - Analytics dashboard.
 - Advanced admin controls.
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Chapter 16 – Conclusion

UniSwap demonstrates the practical application of modern web technologies in solving a real-world campus problem. The project provides a secure, responsive, and scalable marketplace that improves the buying and selling experience for students and staff. By

integrating real-time communication, product management, and user-friendly dashboards, the system creates a trusted environment for campus transactions. Its modular design allows future enhancements, making it suitable for broader institutional adoption.

References

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